

Vinith Chaduvu

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PROFESSIONAL SUMMARY

AI Engineer with production experience developing enterprise ML systems at Accenture and MS in Artificial Intelligence. Expertise in NLP, computer vision, and RAG systems with proven track record of deploying models processing 10K+ daily interactions. Skilled in PyTorch, TensorFlow, and cloud infrastructure with focus on scalable AI solutions.

EXPERIENCE

Packaged App Development Associate <i>Accenture</i>	Oct 2022 – Oct 2023 <i>Bangalore, India</i>
- Built and deployed production sentiment analysis system using NLTK and Python for Vodafone’s customer service platform, processing 10K+ daily customer interactions with 92% classification accuracy - Developed conversational AI chatbot integrated with existing CRM systems, reducing Level 1 support tickets by 35% and improving customer response time by 40% - Optimized SQL queries and implemented database indexing strategies across 3 enterprise applications, achieving 24% performance improvement (2.1s to 1.6s average response time) - Collaborated with cross-functional teams to gather ML requirements and deliver AI solutions within aggressive client timelines	

AI/ML PROJECTS

Adaptive Global LCA Advisor <i>Python, Mistral-7B, Qdrant, RAG, Fine-tuning</i>	2025
- Engineered enterprise-grade RAG system for carbon accounting compliance with fine-tuned Mistral-7B model, achieving 87.2% Precision@3 across 23K+ emission factors from 44 global regions - Implemented high-performance Qdrant vector search with 148ms query latency, enabling real-time EU Carbon Border Adjustment Mechanism compliance queries for multinational corporations - Designed scalable architecture supporting multi-regional carbon footprint calculations with automated regulatory updates	
TaskFin: Conversational Bill Payment System <i>Claude API, LangChain, FastAPI, NLP</i>	2025
- Developed secure fintech application with Claude 3.5 Sonnet API enabling natural language bill payments, achieving 96% intent classification accuracy across diverse payment scenarios - Built robust FastAPI backend with JWT authentication, fuzzy string matching algorithms for account/vendor resolution, and comprehensive error handling for financial transactions - Implemented end-to-end security protocols including PCI DSS compliance measures and real-time fraud detection	
AI-Powered Veterinary Diagnostics <i>PyTorch, ResNet50, Computer Vision, Medical AI</i>	2024
- Automated Vertebral Heart Score estimation from canine X-rays using fine-tuned ResNet50 with custom regression head for precise 6-point anatomical landmark detection - Achieved 4.2% Mean Absolute Percentage Error on VHS calculations, reducing manual veterinary diagnosis time by 80% for cardiomegaly screening while maintaining clinical accuracy standards - Deployed model with confidence scoring system and uncertainty quantification for clinical decision support	

TECHNICAL SKILLS

Programming & ML Frameworks: Python, SQL, PyTorch, TensorFlow, Scikit-learn, OpenCV, NLTK, Hugging Face
AI/ML Specializations: Deep Learning, NLP, Computer Vision, RAG Systems, Model Fine-tuning, MLOps, Deployment
Infrastructure & Tools: AWS (EC2, S3, SageMaker), Docker, Kubernetes, Vector DBs (Qdrant, FAISS), Neo4j
Development: FastAPI, Streamlit, Flask, LangChain, RESTful APIs, Git, MLflow, Weights & Biases

EDUCATION

Yeshiva University, Katz School of Health and Science <i>Master of Science in Artificial Intelligence; GPA: 3.8/4.0</i>	New York, NY <i>May 2025</i>
Vignan Institute of Technology and Sciences <i>Bachelor of Technology in Computer Science</i>	Hyderabad, India <i>July 2022</i>

CERTIFICATIONS

Machine Learning Course - Stanford University, DeepLearning.AI (Coursera)
Deep Learning Specialization - Stanford University, DeepLearning.AI (Coursera)